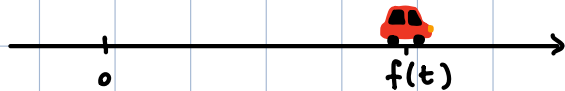


Motivation: average velocity vs instantaneous velocity

Suppose you are driving on a straight road.



$s(t)$ = position at the time t

$v(t)$ = instantaneous velocity

(what you read on speedometer)

Average velocity / rate of change on $[t_1, t_2]$:

$$\begin{aligned} av &= \frac{\Delta s}{\Delta t} = \frac{s(t_2) - s(t_1)}{t_2 - t_1} \\ &= \frac{s(t_1 + h) - s(t_1)}{h} \end{aligned}$$

if $t_2 = t_1 + h$.

Examples: 1) If $s(t) = 7t^2$, approximate $v(1)$.

We can approximate $v(1)$ with the average velocity on a small interval from $t = 1$ to $t = 1 + h$.

Notation: $av(h) = \frac{\Delta s}{\Delta t} = \frac{s(1+h) - s(1)}{h}$

$$\bullet \quad av(0.1) = \frac{7 \cdot (1.1)^2 - 7 \cdot 1^2}{1.1 - 1} = \boxed{14.7}$$

$$\bullet \quad av(-0.1) = \frac{7 \cdot 1^2 - 7 \cdot (0.9)^2}{1 - 0.9} = \boxed{13.3}$$

← approximations
of $v(1)$

2) With $s(t) = 7t^2$ as in 1), use a table of values of the average velocity to estimate $v(1)$.

We find what value $av(h)$ tends to as h gets closer to 0.

h	$av(h)$
0.1	14.7
0.01	14.07
0.001	14.007
0	undef.
-0.001	13.993
-0.01	13.93
-0.1	13.3

As h gets closer to 0, $av(h)$ gets closer to 14.

$$v(1) = \lim_{h \rightarrow 0} av(h) = 14.$$

3) Use algebra to confirm the estimate from 1).

$$\begin{aligned} av(h) &= \frac{s(1+h) - s(1)}{h} = \frac{7(1+h)^2 - 7}{h} \\ &= \frac{7(1+2h+h^2) - 7}{h} = \frac{14h+7h^2}{h} = 14+7h \quad \text{if } h \neq 0. \end{aligned}$$

cannot take $h = 0$ in this expression: simplify algebraically first

Now we can calculate the limit as $h \rightarrow 0$ to find $v(1)$:

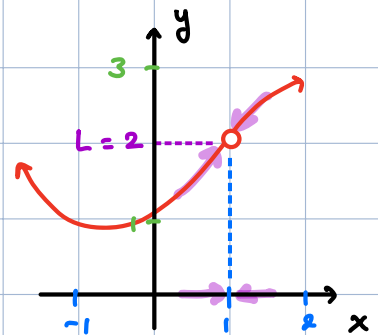
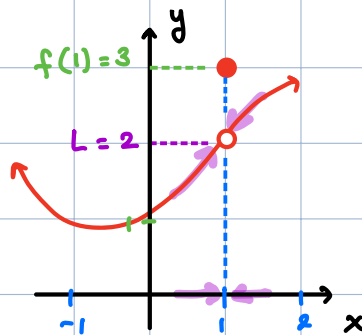
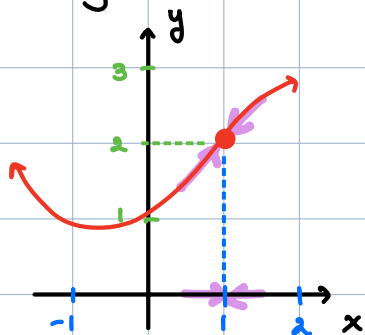
$$v(1) = \lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} = \lim_{h \rightarrow 0} (14+7h) = 14+7 \cdot 0 = \boxed{14}$$

Formal definition of limits:

$\lim_{x \rightarrow a} f(x) = L$ means that $f(x)$ gets arbitrarily close to L when x gets close to a (but $x \neq a$).

↳ "the limit of $f(x)$ as x goes to a is L "

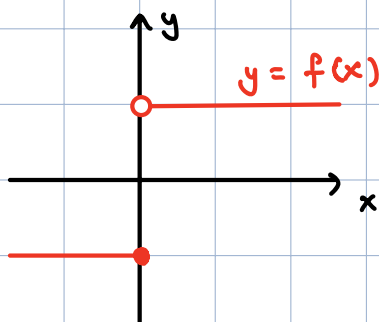
Graphically:



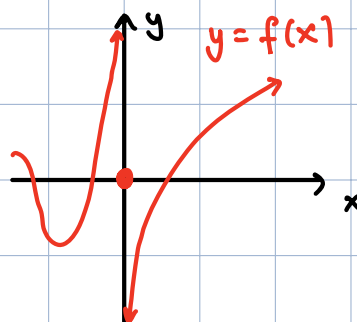
In all 3 cases, $\lim_{x \rightarrow 1} f(x) = 2$.

$\lim_{x \rightarrow a} f(x)$ does not depend on the value of f at a , only on the tendency of $f(x)$ as x gets close to a .

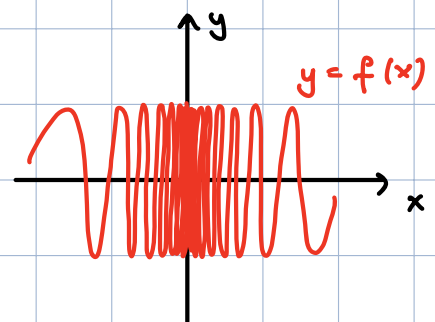
Remark: Limits may or may not exist.



Jump



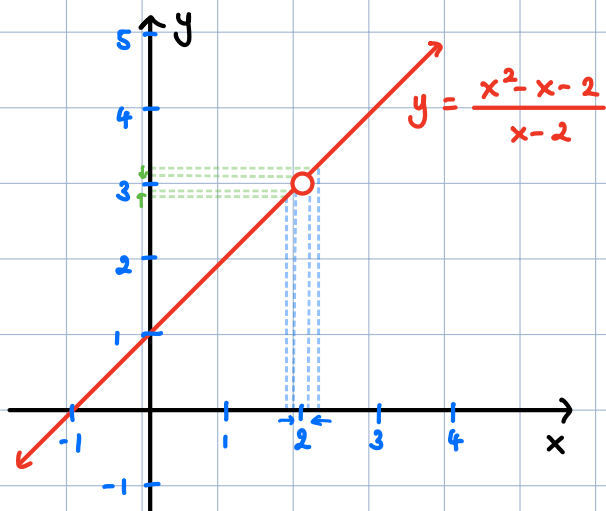
Vertical Asymptote



Infinite oscillation

In all 3 cases, $\lim_{x \rightarrow 0} f(x)$ Does Not Exist (DNE).

Examples : 1) Calculate $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2}$.



$$\frac{x^2 - x - 2}{x - 2} = \frac{(x+1)(x-2)}{x-2} = \begin{cases} x+1 & \text{if } x \neq 2 \\ \text{undef} & \text{if } x = 2 \end{cases}$$

When $x \rightarrow 2$, $y \rightarrow 3$ so

$$\boxed{\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = 3}$$

Formal computation :

$$\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} \quad \text{cannot take } x = 2, \text{ simplify first.}$$

$$= \lim_{\substack{x \rightarrow 2 \\ x \neq 2}} \frac{(x+1)\cancel{(x-2)}}{\cancel{x-2}} = \lim_{x \rightarrow 2} x+1 = \boxed{3}$$

2) Estimate $\lim_{x \rightarrow 8} \frac{x - 8}{\sqrt[3]{x} - 2}$ using a) a table of values, b) a graph.

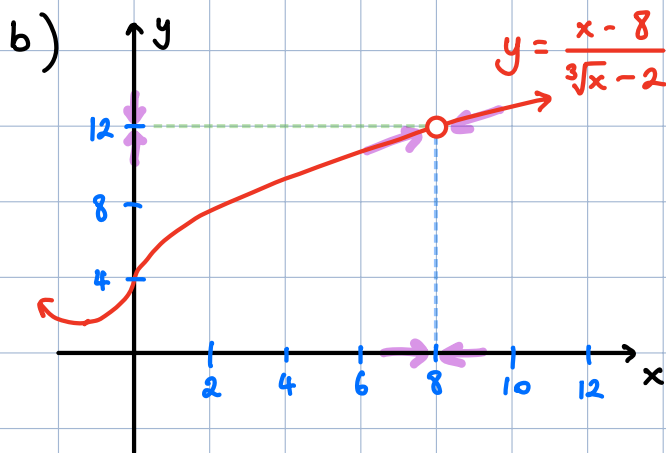
a)

x	$\frac{x-8}{\sqrt[3]{x}-2}$
8.1	12.0498...
8.001	12.0004
8.00001	12.000005...
8	undefined
7.9999	11.99995...
7.999	11.9994...
7.9	11.9498...

As x gets closer to 8,

$\frac{x-8}{\sqrt[3]{x}-2}$ gets closer to 12.

$$\Rightarrow \boxed{\lim_{x \rightarrow 8} \frac{x - 8}{\sqrt[3]{x} - 2} = 12}$$



$$\lim_{x \rightarrow 8} \frac{x-8}{\sqrt[3]{x}-2} = 12$$

3) Suppose that the position of an object moving on an axis is given by $s(t) = 5 - \frac{8}{t}$. Find:

a) the average velocity on $[1, 2]$.

b) the instantaneous velocity at $t = 2$, $v(2)$.

$$\begin{aligned} \text{a) } av &= \frac{\Delta s}{\Delta t} = \frac{s(2) - s(1)}{2 - 1} & s(2) &= 5 - \frac{8}{2} = 1 \\ & & s(1) &= 5 - \frac{8}{1} = -3 \\ &= \frac{1 - (-3)}{2 - 1} = \boxed{4}. \end{aligned}$$

b) To find $v(2)$ we first find and simplify the average velocity $av(h)$ between $t = 2$ and $t = 2 + h$, and then we take the limit as $h \rightarrow 0$.

$$\begin{aligned} av(h) &= \frac{\Delta s}{\Delta t} = \frac{s(2+h) - s(2)}{h} = \frac{\left(5 - \frac{8}{2+h}\right) - 1}{h} \\ &= \frac{4 - \frac{8}{2+h}}{h} = \frac{\frac{4(2+h) - 8}{2+h}}{h} = \frac{8 + 4h - 8}{h(2+h)} = \frac{4h}{h(2+h)} \\ &= \frac{4}{2+h} \quad \text{if } h \neq 0. \end{aligned}$$

$$\text{So } v(2) = \lim_{h \rightarrow 0} av(h) = \lim_{h \rightarrow 0} \frac{4}{2+h} = \frac{4}{2+0} = \boxed{2}.$$

4) Find the instantaneous rate of change of

$$f(x) = \sqrt{x-3} \quad \text{at } x=7.$$

First, we find and simplify the average rate of change $\text{arc}(h)$ between $x=7$ and $x=7+h$.

$$\begin{aligned} \text{arc}(h) &= \frac{\Delta f}{\Delta x} = \frac{f(7+h) - f(7)}{h} = \frac{\sqrt{7+h-3} - \sqrt{7-3}}{h} && \text{cannot take } h=0, \\ & && \text{simplify first.} \\ &= \frac{\sqrt{4+h} - 2}{h} \cdot \frac{\sqrt{4+h} + 2}{\sqrt{4+h} + 2} && \text{multiply by conjugate} \\ &= \frac{(\sqrt{4+h})^2 - 2^2}{h(\sqrt{4+h} + 2)} = \frac{4+h-4}{h(\sqrt{4+h}+2)} = \frac{\cancel{h}}{\cancel{h}(\sqrt{4+h}+2)} = \frac{1}{\sqrt{4+h}+2} \end{aligned}$$

Then, we take the limit as $h \rightarrow 0$ to find the instantaneous rate of change at $x=7$:

$$\lim_{h \rightarrow 0} \text{arc}(h) = \lim_{h \rightarrow 0} \frac{1}{\sqrt{4+h} + 2} = \frac{1}{\sqrt{4+0} + 2} = \boxed{\frac{1}{4}}.$$

5) Find the instantaneous rate of change of

$$f(x) = \frac{6}{3x-4} \quad \text{at } x=2.$$

First, we find and simplify the average rate of change $\text{arc}(h)$ between $x=2$ and $x=2+h$.

$$\text{arc}(h) = \frac{\Delta f}{\Delta x} = \frac{f(2+h) - f(2)}{h} = \frac{\frac{6}{3(2+h)-4} - 3}{h} = \frac{\frac{6}{2+3h} - 3}{h}$$

$$= \frac{6 - 3(2+3h)}{h(2+3h)} = \frac{6 - 6 - 9h}{h(2+3h)} = \frac{-9\cancel{h}}{\cancel{h}(2+3h)} = \frac{-9}{2+3h}$$

Then, we take the limit as $h \rightarrow 0$ to find the instantaneous rate of change at $x = 2$:

$$\lim_{h \rightarrow 0} \text{arc}(h) = \lim_{h \rightarrow 0} \frac{-9}{2+3h} = \frac{-9}{2+3 \cdot 0} = \boxed{-\frac{9}{2}}.$$